

Trade Study TS-001

Portable Power System Selection for Ground Control Station

Project: Instrument Case Modernization for GCS as a subsystem of TOMTOM

Prepared By: Serena Green

Date: July 11, 2026

Version: 002

| Version | Date        | Author       | Description of Change                                                                                                                                                                                                                                |
|---------|-------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 11 Jul 2026 | Serena Green | Initial draft created. Defined objectives, design requirements, alternatives, evaluation criteria, and preliminary decision matrix.                                                                                                                  |
| 0.2     | 16 Jul 2026 | Serena Green | Updated evaluation based on final hardware procurement. Selected 12.8 V LiFePO <sub>4</sub> battery as the preferred portable power architecture. Revised weighted scores, discussion, and decision to reflect the implemented system configuration. |
| 1.0     | TBD         | Serena Green | Baseline release for the TOMTOM Ground Control Station prototype. Document approved for project documentation and configuration baseline.                                                                                                            |

## 1. Objective

The objective of this trade study is to evaluate candidate portable power architectures for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM) Ground Control Station (GCS) and select the solution that best satisfies the system's operational, technical, and cost requirements. The GCS is being developed as a supporting subsystem of the TOMTOM ISR demonstrator and is responsible for mission planning, vehicle command and control, telemetry monitoring, payload management, and operator situational awareness.

The GCS hardware is being constructed by repurposing the internal components of an ASUS G73S laptop and integrating them into a vintage portable instrument case to create a rugged, portable field computing platform. This custom enclosure is intended to house the laptop motherboard, display, power distribution system, communications equipment, and future mission support hardware while providing improved portability and field usability compared to the original laptop configuration.

Because the Ground Control Station is a subsystem of the larger TOMTOM system, its design must remain consistent with the operational objectives, system requirements, and architectural constraints established for TOMTOM. The selected power solution must therefore provide sufficient endurance for field operations, support future expansion of mission equipment, enable reliable operation under portable deployment conditions, and maintain compatibility with planned system enhancements such as integrated telemetry radios, external peripherals, and renewable energy charging capabilities. In addition to meeting technical performance objectives, the selected architecture should provide an appropriate balance between cost, reliability, maintainability, portability, and future scalability while supporting the overall development goals of the TOMTOM project.

### Assumptions

- The ASUS G73S motherboard remains the primary computing platform.
- The GCS will operate independently of commercial power for a minimum of two hours.
- The power system will support future integration of telemetry radios and additional mission equipment.
- The design will be installed within the selected portable instrument case.

### Constraints

- Prototype budget shall not exceed \$200 for the initial power subsystem.

- The architecture shall preserve compatibility with future solar charging.
- Components shall be commercially available.

## 2. Design Requirements

| Requirement              | Value               |
|--------------------------|---------------------|
| Runtime                  | $\geq 2$ hours      |
| Operating Voltage        | 19 VDC laptop input |
| Budget                   | <\$200 preferred    |
| Rechargeable             | Yes                 |
| Field Portable           | Yes                 |
| Future Solar Integration | Desired             |
| Safe Operation           | Required            |

## 3. Alternatives Considered

### Alternative A

OEM ASUS Laptop Battery

Advantages

- Integrated charging
- No additional electronics
- Lightweight

Disadvantages

- Currently does not work and would necessitate purchasing a replacement to be initially considered
- Limited runtime
- Difficult to replace in custom enclosure
- No solar charging capability

### Alternative B

## 12V Sealed Lead Acid (SLA)

### Advantages

- Lowest cost
- Widely available
- Simple charging
- High current capability
- Rugged and durable

### Disadvantages

- Heavy
- Lower cycle life
- Lower energy density

## **Alternative C**

### 12.8V LiFePO<sub>4</sub> Battery

### Advantages

- Comparably lightweight
- Long cycle life
- Excellent energy density
- Solar compatible

### Disadvantages

- Highest purchase cost

## **Alternative D**

### Commercial Portable Power Station

### Advantages

- Plug-and-play
- Built-in inverter
- Built-in charging
- Safe

### Disadvantages

- Expensive
- Larger footprint
- Less customizable

## 4. Evaluation Criteria

| Criterion           | Weight |
|---------------------|--------|
| Initial Cost        | 25%    |
| Runtime             | 20%    |
| Weight              | 15%    |
| Reliability         | 15%    |
| Expandability       | 10%    |
| Ease of Integration | 10%    |
| Maintainability     | 5%     |

The evaluation criteria were weighted according to project priorities. Initial acquisition cost received the highest weighting (25%) because minimizing prototype cost was identified as the primary design driver. Runtime (20%) and reliability (15%) were prioritized to ensure field usability. Weight, expandability, ease of integration, and maintainability were assigned lower weights because they represent desirable characteristics that can be improved through future design iterations.

## 5. Decision Matrix

| Criteria        | Weight | OEM Battery | SLA | LiFePO <sub>4</sub> | Power Station |
|-----------------|--------|-------------|-----|---------------------|---------------|
| Cost            | 0.25   | 3           | 5   | 2                   | 2             |
| Runtime         | 0.20   | 2           | 4   | 5                   | 5             |
| Weight          | 0.15   | 5           | 2   | 5                   | 3             |
| Reliability     | 0.15   | 4           | 5   | 5                   | 5             |
| Expandability   | 0.10   | 2           | 4   | 5                   | 4             |
| Integration     | 0.10   | 5           | 4   | 4                   | 5             |
| Maintainability | 0.05   | 3           | 5   | 5                   | 5             |

Alternatives were evaluated using a five-point ordinal scale, where 1 represents poor compliance with the criterion and 5 represents excellent compliance.

## 6. Weighted Scores

| Alternative         | Total Score |
|---------------------|-------------|
| OEM Battery         | 3.15        |
| SLA                 | 4.15        |
| LiFePO <sub>4</sub> | 4.55        |
| Power Station       | 3.75        |

## 7. Discussion

The 12.8V LiFePO<sub>4</sub> battery achieved the highest overall weighted score and was selected as the preferred portable power architecture for the TOMTOM Ground Control Station. Although the initial acquisition cost is higher than a comparable Sealed Lead Acid (SLA) battery, the LiFePO<sub>4</sub> solution provides significant operational advantages that better support the long-term objectives of the project.

The selected battery offers substantially lower weight, higher energy density, longer cycle life, and improved portability while satisfying the required minimum two hour operational endurance. These characteristics are particularly important for a portable field system intended to support mission planning, command and control, telemetry monitoring, and future payload expansion. The reduced system weight also improves transportability and decreases operator fatigue during deployment.

The selected architecture remains compatible with future system enhancements, including solar charging capability, additional mission electronics, and expanded communications equipment. While the higher initial purchase cost represented a greater upfront investment, the improved service life, reduced maintenance requirements, and superior electrical performance provide greater lifecycle value than the lower-cost SLA alternative.

As the project evolved, additional funding became available for the power subsystem, allowing selection of the LiFePO<sub>4</sub> battery without exceeding the overall project budget. This decision better aligns the prototype with the operational goals established for the TOMTOM Ground Control Station while providing a robust foundation for future system development.

## 8. Decision

**Selected Alternative: 12.8V LiFePO<sub>4</sub> Battery**

**Rationale**

The LiFePO<sub>4</sub> battery provides the best overall balance of:

- Lightweight portable operation
- Long operational life
- High energy density
- Excellent reliability
- Solar charging compatibility
- Future expandability
- Low maintenance requirements

Although the initial acquisition cost is greater than the other evaluated alternatives, the improved performance, reduced lifecycle cost, and compatibility with future system expansion make the LiFePO<sub>4</sub> battery the most appropriate power solution for the TOMTOM Ground Control Station prototype. During the initial design phase, the Sealed Lead Acid (SLA) battery was selected because it provided the highest value within the project's budget constraints while satisfying all operational requirements. Following subsequent hardware procurement and an increase in the available project budget, the power subsystem was reevaluated. The improved energy density, reduced weight, longer service life, and enhanced portability of the LiFePO<sub>4</sub> battery justified revising the original design decision. This change represents an engineering refinement based on evolving project constraints rather than a correction of the original analysis.



## Appendix A: Design Evolution

### **Revision 001**

- Selected SLA battery due to budget limitations.

### **Revision 002**

- Project funding increased.
- Procured 12.8V LiFePO<sub>4</sub> battery.
- Updated power architecture.
- Reduced system weight.
- Improved energy density.
- Increased expected service life.